1. Consider a system with a two-component vector order parameter $\vec{m}(\vec{x})$ whose free energy expansion does not have the quartic term (tricritical point).

$$\beta L = \int d^d r [\frac{\gamma}{2}(\nabla \vec{m})^2 + \frac{1}{2} a_2 \vec{m} \cdot \vec{m} + v(\vec{m} \cdot \vec{m})^3, \text{with } a_2 = a(T - T_c), \& \ v > 0, a > 0$$

Repeat some of the calculations for this system. [Kardar Chapter 3, Problem 4].

(a) Calculate the heat capacity singularity as $T \to T_c$ in the saddle point approximation.

(b) Include both longitudinal and transverse fluctuations by setting

$$\vec{m}(\vec{x}) = [\overline{m} + \phi_1(\vec{x})]\hat{e}_1 + \phi_2(\vec{x})\hat{e}_2,$$

and expanding the free energy to to quadratic order in ϕ. Here $\overline{m}\hat{e}_1$ is the mean-field order parameter. Get the expression for the Landau-Ginzburg free energy in this Gaussian approximation below and above T_c.

(c) Calculate the longitudinal and transverse correlation functions. You can directly invoke the equipartition theorem here without needing to evaluate the thermodymanic averages.
[Similar to what is done on p341 & p179 Goldenfeld].

(d) Compute the first correction to the saddle point (Gibbs) free energy from fluctuations, viz., do the functional integral.
[Similar to p178 Goldenfeld].

(e) Find the fluctuation correction to the heat capacity above and below the transition.